Sensation
Uninterpreted sensory impressions created by detection of a stimulus

Perception
Processes to make sense of sensations

Recognition
Identification of stimuli as objects we know

Object recognition

Main path
Detailed concepts

Definition

Basic idea

Incoming visual information is matched with **internal object representations** held in object stores (memory)

> If incoming visual information matches an object representation → object is recognised
> If incoming information does not match an object representation → object is not recognised, re-analysis

3 theories of how matching is achieved

Structural analysis
Multiple sequential **levels of representations** are processed

This means

Objects may differ on initial levels, but become equivalent on later stages

Most prominent advocate

Template matching
Object representations are **whole object templates** (e.g., A)

This means

Incoming information undergoes **visual transformations** (e.g., rotation, size scaling) until template match is achieved

- **Problem**: Requires large number of templates

Feature analysis
Object representations are **lists of object features** (e.g., / _ \)

This means

Different features (e.g., an additional line) discriminate objects (e.g., P and R)

- **Problem**: Spatial arrangements of features are not considered

David Marr (1982)
The visual system proceeds through **4 levels of representations**

1. Raw primal sketch
2. Full primal sketch
3. 2½-D sketch
4. 3-D model

Definitions

Raw primal sketch
Intensity (brightness) changes across the retina

Full primal sketch
Geometrical organisation, defining edges and contours

2½-D sketch (**viewpoint-dependent** object representation)
Spatial locations of visible surfaces, including depth

3-D model (**standard object representation**)
Real shapes and surfaces, their relative positions to each other

Inspired a cognitive model of object recognition

Ellis & Young (1988)

Model

Legend
- Functional processing stage
- → Flow of information
- Level of processing
- ~ Corresponds to

Stimulus (object)

~ Sensation

Initial representation

~ Primal sketches

~ Perception

Viewer-centred representation

~ 2½-D sketch

Object-centred representation

~ 3-D model

Object recognition units

~ Recognition

Semantic system

Name retrieval

~ Visual processing

~ Semantic processing

Helps

Classifying different types of **impairments**

Such as

Agnosia
= Inability to recognise seen objects

- **Apperceptive agnosia**

Impaired **perceptual processing** stages (**viewer-centred representations**)
Impaired **shape perception**
Cannot copy and match simple forms
e.g., Mr S

- **Associative agnosia**

Impaired **semantic processing** stages (**object recognition units, semantic system**)
Can copy and match simple forms, but intact **percept has no meaning**
e.g., AB

Such as

Perceptual categorisation deficit
= Inability to form **object-centred representations**

Can recognise objects from **usual views**, cannot recognise objects from **unusual views**
e.g., patients with right posterior lesions